

# 1. Problem Analysis Report

## The Root Cause: Reactive Logistics in a Predictive Era

Despite India facing recurring and predictable natural disasters (monsoons in Assam/Bihar, cyclones in Odisha/West Bengal), the disaster response supply chain remains fundamentally **reactive**.

NGOs and District Disaster Management Authorities (DDMAs) typically mobilize resources *after* a disaster strikes and media coverage highlights the severity. This lag results in a “golden window” failure—the critical first 48-72 hours where intervention saves the most lives.

While rich open datasets exist (IMD weather models, ISRO satellite data, NDMA incident reports), they are heavily siloed. The root cause of supply chain failure during disasters is not a lack of data, but the **lack of operational data fusion and predictive logistics**.

## Deep Dive: The Cost of Chaos

During a severe cyclone or flood, communication lines break. Central NGO hubs (e.g., in Bhubaneswar) have limited visibility into localized ground realities.

1. **Misallocation:** Without data, NGOs estimate supplies based on historical biases. High-visibility areas receive redundant supplies from multiple organizations, while remote districts face severe deficits.
2. **Logistical Bottlenecks:** Trucks are dispatched without optimized routing, leading to delays caused by inundated roads or excessive fuel costs.

## Market Trends & User Friction

- **Market Trend:** The frequency of extreme weather events is rising by 20-30% annually due to climate change. Simultaneously, government policies are pushing for tech-enabled disaster resilience.
- **User Friction:** NGO Operations Managers currently stitch together Excel sheets, WhatsApp updates, and TV news to make life-or-death inventory decisions. They experience immense cognitive overload and high friction when trying to calculate dispatch quantities manually. Nirvana solves this by turning siloed data into actionable, exact dispatch manifests.